



Plastic-eating enzyme

Scientists have accidentally created a mutant enzyme that eats plastic bottles. It could solve plastic pollution. <https://digit.in/mutenzyme>



Nuclear power in space

NASA to announce project 'kilopower' which will use a nuclear reactor to produce clean energy on the Moon. <https://digit.in/kilopower>

gas industry was gaining speed. It was noticed that the flow of gas was being blocked by these mysterious structural formations. This led to investment in research work to impede the formation of gas hydrates in pipelines. These pipelines created a natural environment for the hydrates to form at low temperatures and high pressures.

In the 1960s, during the ongoing exploration and production of gas wells in the Messoyahka gas field of the Western Siberian basin, methane hydrates were first encountered occurring in a natural state followed by several other global sightings. Obtaining core samples is a regular feature of any drilling program in oil and gas explorations. These samples are usually rocks dug out of the earth in tubes. During a deep sea drilling program off the coast of Guatemala, the core samples recovered showed presence of hydrate bearing rocks. This discovery was the dawn of viewing them as a natural methane source and the possibility that these are a widespread occurrence in nature

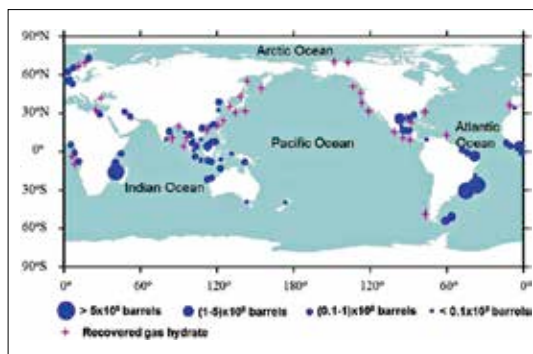
METHANE HYDRATES AS AN ENERGY SOURCE

Methane is a widely used source of fuel and is relatively clean. It is also a large component of natural gas. But why pick methane hydrates as an energy source? They have great potential to be optimised and monetised. At normal surface temperature and pressure, one cubic foot of solid methane hydrate will release about 164 cubic feet of methane gas. It has been estimated that globally, reserves in place would be about 700,000 trillion cubic feet (Tcf) of methane within hydrates.

Having said that, mind you, not all of it is economically feasible for recovery. Picture it as a pyramid. The recoverable varieties are found at the top of the pyramid structure. Hundreds of Tcf has been estimated to be found in the arctic sandstones. While the marine sands seem to hold 10,000s of Tcf of methane. The lower you travel down the pyramid the harder it is to make economically feasible recovery.

100,000s of Tcf are located in this part of the pyramid. Each of these deposits are in a different geological setting and have their own recovery challenges. Several oil and gas majors have taken a keen interest in bringing methane to surface level. From an international standpoint, countries like the USA, China and Japan are voraciously looking into methane hydrates as a fuel source. Japan is one of those countries that are ecstatic to invest in this. As a country with high fuel import costs, they've invested excellent research and were the first ones to successfully produce methane from hydrates.

In our own backyard, India is showing keen interest in the commercial production of hydrates. India has set aside about \$56 million for research alone. This has been an ongoing area



Map of distribution of Hydrate presence around the world

of interest, in fact, since 1996. Several leading organizations are working towards defining the zones of interest in the continental shelf regions of both, the Bay of Bengal and the Arabian Sea. Research is also indicative of gas hydrate deposit off the coasts of the Krishna-Godavari river deltas and the Andaman & Nicobar Islands.

UNDERSTANDING ITS EFFECTS ON THE CLIMATE

As a civilization going through a major cycle of climate change, we are asked to assess environmental impact. This topic will surely perk the interests of our climate change enthusiasts and so it should. Methane is widely known as a greenhouse gas and is perfectly capable of affecting Earth's carbon cycle.

Much research needs to be completed before we go "El Dorado" on hydrates. A few key questions to ask regarding hydrates would be as follows:

- Do we know when to stop? We can't be greedy and turn an exploration mission into an exploitation mission. That won't turn out well.
- Are we well equipped to deal with introducing additional methane into our environmental system?

Minimal environmental impact and maximum economic recovery is an equilibrium that should be reached after careful research and considerations before we move ahead.

WHAT DO WE THINK SO FAR?

Yes, it is fascinating and sounds like the answer to several problems that comes with the paucity of non-renewable fuels. Hell, if someone told us there exists ice that we can set on fire and they exist in rock under the sea, it would seem like a fairy tale. As we venture into newer possibilities of fuel sources, it is key to have a thorough understanding of its repercussions, its pros and its cons. If used correctly, they are the answer to our energy problems

as they are plentiful, abundant and easy to mine. There possibly is more methane present in hydrates than in all petroleum reserves globally. It's no secret that the oil industry has been on the decline for the last three years due to several (geopolitical) reasons. Gas hydrate production is a realm of unconventional energy potential which has brought together several big powers to collectively invest in research into exploring, developing and producing hydrate fields. Once a nuisance in the oil industry, hydrates might now actually be the prophet that leads an exodus of transferable skills (geologists, petroleum engineers, etc.) into the happily ever after.

Written with inputs from Andrea Dsilva, Geoscientist and oil industry specialist